PVDIS PAC Study

Cumulative data, diagnostics, and reproducibility record

PRIMARY GOAL: Inelastic/resonance A_PV for preliminary MOLLER workshop results by Monday afternoon October 26; more complete results Tuesday October 27.

Study opened October 25, 2026. This document grows by adding validated run snapshots; earlier run files are retained with their own hashes.

Scope	Current status
Executed physics	Fresh 11 GeV scan: 896 points, 10,800 controlled response cases, 40 landmarks; four published resonance means. Earlier 804-point W scans are retained as historical evidence.
Construction	Light-valence, charge-symmetric rotation of electromagnetic F1F2IN21; explicitly assumed weak/nuclear response.
Validated locally	Pinned input identities, repeated raw output, algebra and numerical CSV consistency; see per-run checks.
Not established	Detector acceptance, radiative migration, full weak resonance model, validated rates, parameter fit, or optimal PAC settings.
Exposure	30 PAC days total; sequential configurations partition this total; simultaneous spectrometers share exposure. Not used in this benchmark.

Run register

Capture UTC	Run / mode	Gates
2026-10-25T18:05:51.821390+00:00	First resonance benchmark; collect_retained_evidence	11 PASS
2026-10-25T18:10:24.773791+00:00	Reproducible resonance benchmark - report baseline; full_fortran_run	11 PASS

Current physics study: resonance_studies/20261025T193128315988Z. Its separate manifest and diagnostics appear with the new result; the register above records the historical benchmark captures.

Reading order: fresh 11 GeV inelastic/resonance result and response constraints; historical benchmarks; received fit-notebook audit; machine-data dictionary; reproduction and references.

PRIMARY RESULT: inelastic/resonance A_PV

Fresh study ID: 20261025T193128315988Z

The immediate goal is inelastic/resonance A_PV for preliminary MOLLER workshop results Monday afternoon October 26, with more complete work Tuesday October 27. Coupling-ellipse optimization is secondary. The earlier point benchmark and notebook audit do not constitute completion of this goal.

New calculation	Executed scope
11 GeV coverage	896 fresh model points at eight angles from 12.5 to 30 degrees, W=1.10-2.20 GeV.
Response probes	10800 combinations of axial and longitudinal response assumptions, including the four published means.
Reference kinematics	40 angle/W landmarks, including Delta-region and higher-mass points; not acceptance-integrated or optimized settings.
Reproducibility	Pinned electromagnetic source/tables; fresh compilation; repeated model execution; direct-response algebra checks; per-point JSON and CSV.
Statistical reach	Unavailable until LD2 rates and exposure normalization are verified. Total planning budget remains 30 PAC days.
Scientific status	Conditional light-valence reference curves. Weak resonance amplitudes, resonance/background isospin, nuclear weak response and radiative/acceptance folding remain unresolved.

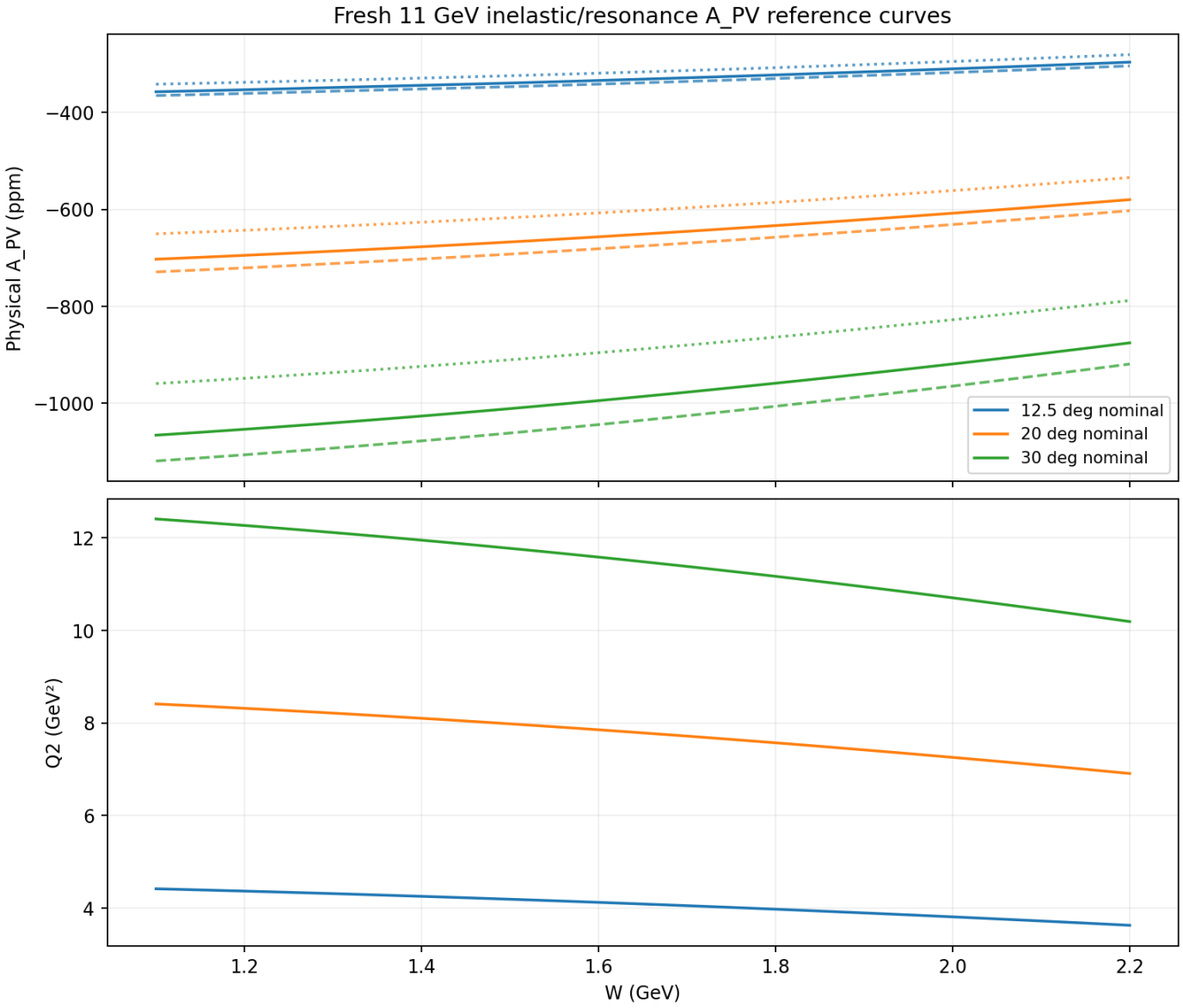
What these calculations resolve

We now separate the vector and axial asymmetry terms, vary the unknown F3 and weak longitudinal responses explicitly, and calculate the 11 GeV kinematics. No data were fitted to choose the response multipliers. Scenario differences are not a probability band or calibrated theory uncertainty.

What they do not resolve

Electromagnetic F1 and F2 do not uniquely fix the weak resonance response. In particular, the smooth isoscalar light-valence transverse ratio differs from the pure isospin-3/2 transition limit. A complete physical prediction requires resonance/background information beyond the current construction.

Fresh 11 GeV curves and axial-response dependence



Solid: nominal light-valence closure; dotted: F3=0; dashed: F3 x 1.5.
Controlled input scenarios, not an uncertainty band; no acceptance or radiative folding.

The plotted examples are fixed scattering angles with Q2 changing along W. The 896-row grid includes all eight angles. Solid curves use the nominal light-valence common-response closure; dotted curves remove F3; dashed curves multiply F3 by 1.5. RgammaZ=Rgamma in this figure.

These curves describe the physical asymmetry before beam-polarization dilution. They are not measured asymmetries, accepted yields, or validated hardware settings. No statistical error is assigned from unverified rates.

Representative 11 GeV inelastic/resonance kinematics

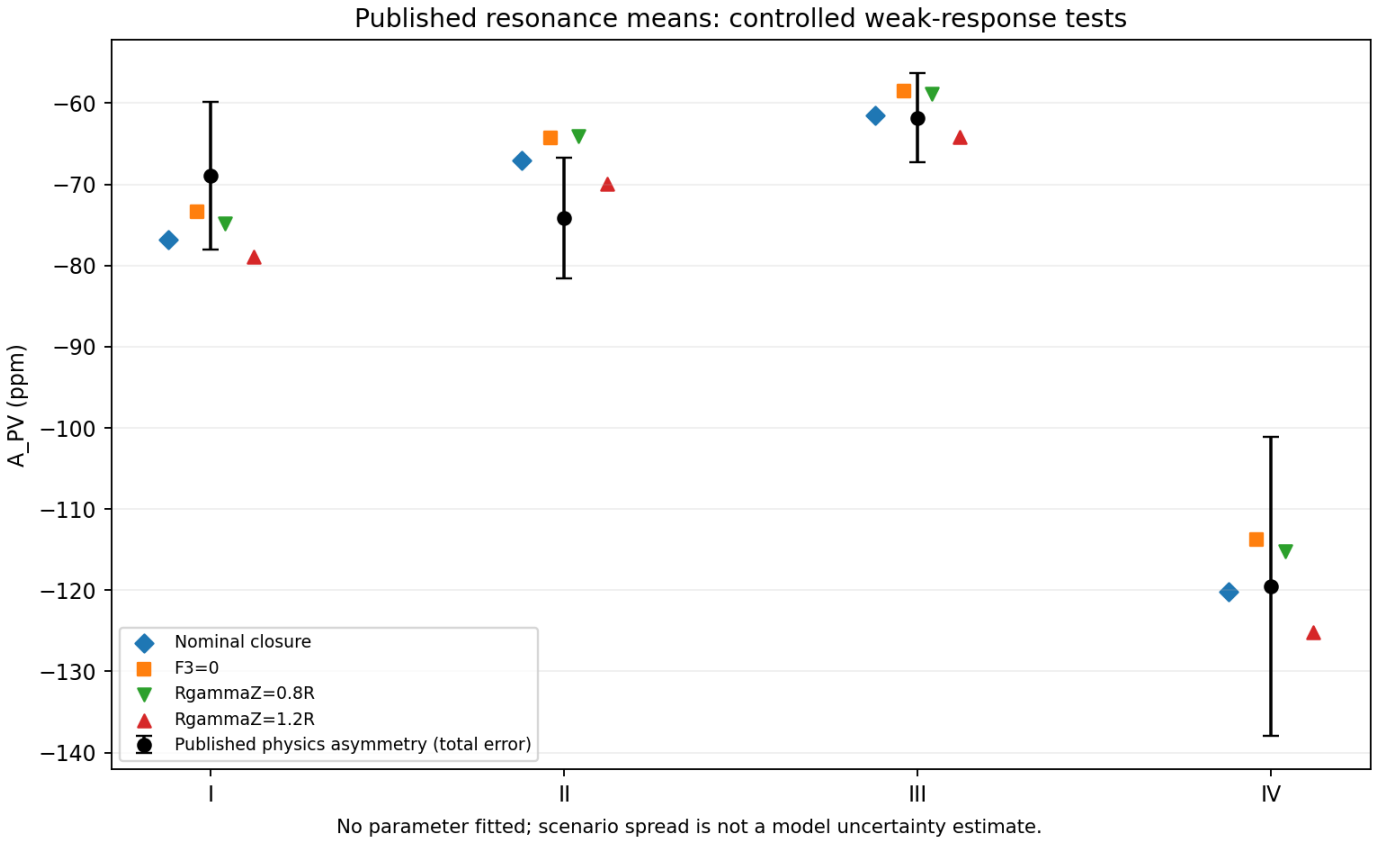
A selected subset of the 40 landmarks is shown below. All values use the declared nucleon kinematics and model assumptions. Momentum availability, magnet configuration, mechanical constraints and target acceptance have not been validated.

Angle (deg)	W (GeV)	Eprime (GeV)	Q2 (GeV2)	Nominal A (ppm)	Axial share (%)
12.5	1.232	8.342	4.350	-351.50	4.34
12.5	1.520	8.012	4.178	-338.05	4.47
12.5	1.680	7.798	4.067	-329.34	4.56
12.5	1.900	7.470	3.895	-316.11	4.75
12.5	2.100	7.136	3.721	-302.99	5.07
20.0	1.232	6.245	8.286	-691.97	7.45
20.0	1.520	5.998	7.957	-664.98	7.51
20.0	1.680	5.838	7.745	-647.41	7.53
20.0	1.900	5.592	7.419	-620.77	7.62
20.0	2.100	5.342	7.088	-594.01	7.77
30.0	1.232	4.147	12.223	-1049.54	9.98
30.0	1.520	3.983	11.738	-1007.75	9.97
30.0	1.680	3.876	11.426	-980.57	9.94
30.0	1.900	3.713	10.944	-939.18	9.93
30.0	2.100	3.547	10.456	-897.57	9.96

At W=1.232 GeV, the nominal physical asymmetry is approximately -351.50 ppm at 12.5 degrees, -691.97 ppm at 20 degrees, and -1049.54 ppm at 30 degrees. The assumed axial term contributes about 4.34%, 7.45%, and 9.98% of the magnitude, respectively.

Files: representative_kinematics.csv, nominal_predictions.csv, response_scenarios.csv, diagnostics.json and data_dictionary.json in this sealed resonance-study directory. Larger asymmetry alone does not establish better statistical reach; rate falls and acceptance must be considered separately.

Published resonance data: response-assumption tests



Setting	Nominal A	F3=0 A	Axial half shift	Rgz 0.8R shift	Rgz 1.2R shift
I	-76.857	-73.430	+1.714	+2.082	-2.082
II	-67.018	-64.232	+1.393	+2.930	-2.930
III	-61.463	-58.512	+1.476	+2.671	-2.671
IV	-120.195	-113.778	+3.209	+5.022	-5.022

All table values are ppm. Removing the assumed axial contribution changes the four means by roughly 2.79-6.42 ppm. A 20% longitudinal-ratio change shifts them by roughly 2.08-5.02 ppm. Such changes can be comparable to experimental precision, and the old agreement does not isolate which weak-response assumptions are correct.

Resonance isospin: a missing physical constraint

In the light-valence isoscalar construction, F1gammaZ/F1gamma=(6/5)(2gVu-gVd)=0.86 at sin2(thetaW)=0.235. A pure isospin-3/2 nucleon transition instead has vector-response ratio 2-4*sin2(thetaW)=1.06. The latter follows from a purely isovector transition; see Gorchtein et al., arXiv:1102.3910, Appendix A after Eq. (A9).

The following values are component-limit diagnostics only. Applying the pure-transition ratio to the entire inclusive electromagnetic strength is not a resonance/background mixture calculation and is not adopted as the nominal inclusive prediction.

W (GeV)	Angle (deg)	Q2 (GeV2)	Valence vector only	Pure I=3/2 vector only
1.263	12.71	0.950	-73.43	-90.51
1.232	12.50	4.350	-336.25	-414.45
1.232	20.00	8.286	-640.43	-789.37
1.232	30.00	12.223	-944.75	-1164.46

This is a 23.26% change in the vector response ratio before specifying any axial amplitude. At the first published mean it moves the vector-only diagnostic from -73.43 to -90.51 ppm. This establishes a model-construction limitation, not a calibrated prediction error for inclusive deuterium.

Next scientific gate

Separate resonant and nonresonant electromagnetic strengths and supply their weak rotations, including relevant proton/neutron transition-amplitude information. Establish the axial response and weak longitudinal ratio. Carlson and Rislow, arXiv:1201.3323v2, describe why resonance helicity-amplitude content matters. Until those ingredients are supplied, retain the 11 GeV curves as conditional references.

Reproduction

```
python tools/run_resonance_study.py
python tools/verify_resonance_study.py --recompute
python tools/build_study.py --mode report
```

The first command performs a fresh local model build/run and creates a new sealed study. The second reconstructs 900 nominal points and 10,800 scenarios from its retained source and raw values. Existing resonance and notebook snapshots are preserved.

Run results: First resonance benchmark

Run ID: 20261025T180551821390Z

Collection mode: collect_retained_evidence. This collection did not rerun Fortran. The original raw values are retained and independently checked against the saved analysis.

Setting	E (GeV)	Q2 (GeV2)	W (GeV)	Data A (ppm)	Stat.	Syst.	Total
I	4.867	0.950	1.263	-68.97	8.47	3.30	9.09
II	4.867	0.831	1.591	-74.12	6.87	2.84	7.43
III	4.867	0.757	1.857	-61.80	5.08	2.11	5.50
IV	6.067	1.472	1.981	-119.56	17.54	5.62	18.42

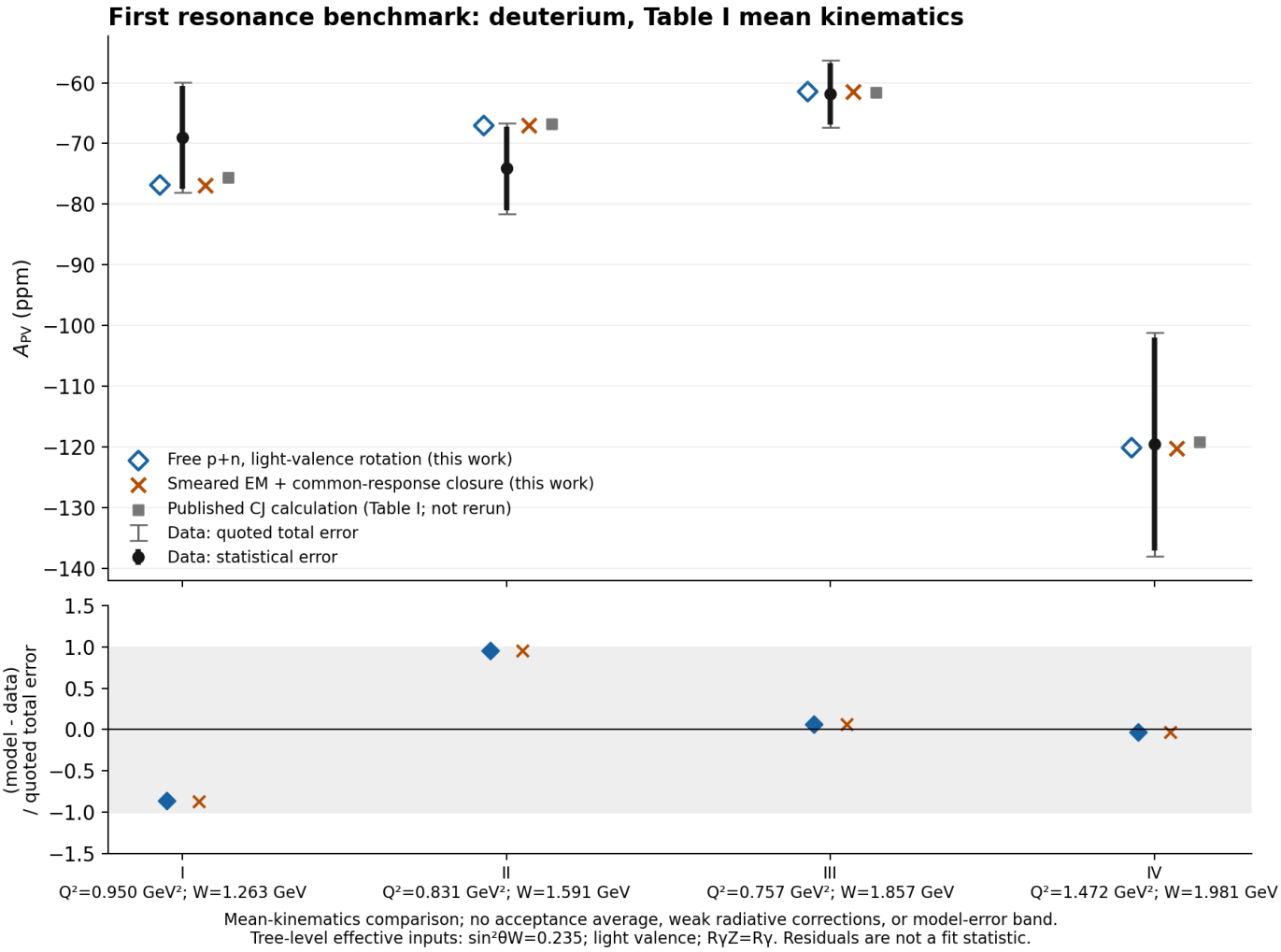
Setting	Free p+n (ppm)	Closure (ppm)	Closure - data (ppm)	Residual / total
I	-76.836	-76.857	-7.887	-0.868
II	-67.039	-67.018	7.102	0.956
III	-61.444	-61.463	0.337	0.061
IV	-120.131	-120.195	-0.635	-0.034

All four free-p+n differences are within 0.96 of one quoted total experimental error. This is a preliminary compatibility check at mean kinematics. It is not an acceptance-averaged comparison, p-value, or proof of model accuracy.

Measured values and their published total errors are transcribed from Wang et al., arXiv:1304.7741v2, Table I [1]. Statistical and systematic columns remain separate in the CSV. No fitted parameter or additional theory error has been used to force agreement.

Files: results/resonance_benchmark.csv contains all four means and diagnostics; results/resonance_W_scan.csv contains the 804 scan rows. The original measured input table is in data/resonance_table_I.csv.

Measured and calculated asymmetries



Black points: physics asymmetries, with statistical and total experimental errors. Blue diamonds: this study's free-p+n rotation. Orange crosses: this study's smeared common-response closure. Gray squares: the paper's published CJ results, included as references and not recomputed.

The residual panel divides model minus data by the quoted total experimental error. It is a diagnostic display, not a combined fit statistic. The missing covariance, acceptance average, and theory uncertainty prevent a goodness-of-fit claim.

Kinematics and electromagnetic diagnostics

These are calculations at each published mean Q2 and W, with the quoted beam energy. The implied angle is calculated from these means; it should not be confused with the central spectrometer setting.

Setting	x	y	Angle (deg)	Y3 free	Y3 closure
I	0.57063	0.18228	12.7149	0.36932	0.37164
II	0.33482	0.27175	12.6008	0.34805	0.34546
III	0.22766	0.36407	12.8712	0.39903	0.40166
IV	0.32595	0.39666	14.7921	0.44465	0.44913

Setting	F1 p+n	F1 d	F1 change (%)	R free	R d
I	0.561875	0.531351	-5.433	0.18034	0.17268
II	0.733047	0.769094	4.917	0.30753	0.31790
III	1.116983	1.127500	0.942	0.34404	0.33441
IV	0.727866	0.724939	-0.402	0.34534	0.33028

Setting	Coefficient C1 (ppm)	Coefficient C2 (ppm)	Closure - free (ppm)
I	102.460313	38.078730	-0.021434
II	89.625811	30.961969	0.020894
III	81.644692	32.793054	-0.019271
IV	158.759559	71.303673	-0.063961

Structure functions are dimensionless. F1 d and F2 d are per-deuteron electromagnetic quantities; the free reference is the sum of proton and neutron. The deuteron call uses CD-Bonn on-shell smearing, explicit GETFY initialization, and no quasielastic contribution.

The small closure-minus-free asymmetry is largely imposed by the shared response assumption. It is not a measured or independently derived bound on weak nuclear effects.

Physics prescription and conventions

Quantity	Value / definition
Signed A	(sigma_R - sigma_L)/(sigma_R + sigma_L)
GF; alpha; M	1.166e-05 GeV^-2; 0.0073; 0.938272 GeV
sin2(thetaW)	0.235; retained effective tree-level baseline
C1; C2	2C1u-C1d = -0.7166667; 2C2u-C2d = -0.09
Couplings	gAe=-1/2; gVe=-1/2+2s2; gVu=1/2-4s2/3; gVd=-1/2+2s2/3; Ciq=2*g_e*g_q
Flavor assumption	Light valence only; charge symmetry; qminus=qplus; no independently supplied sea or axial resonance amplitudes.

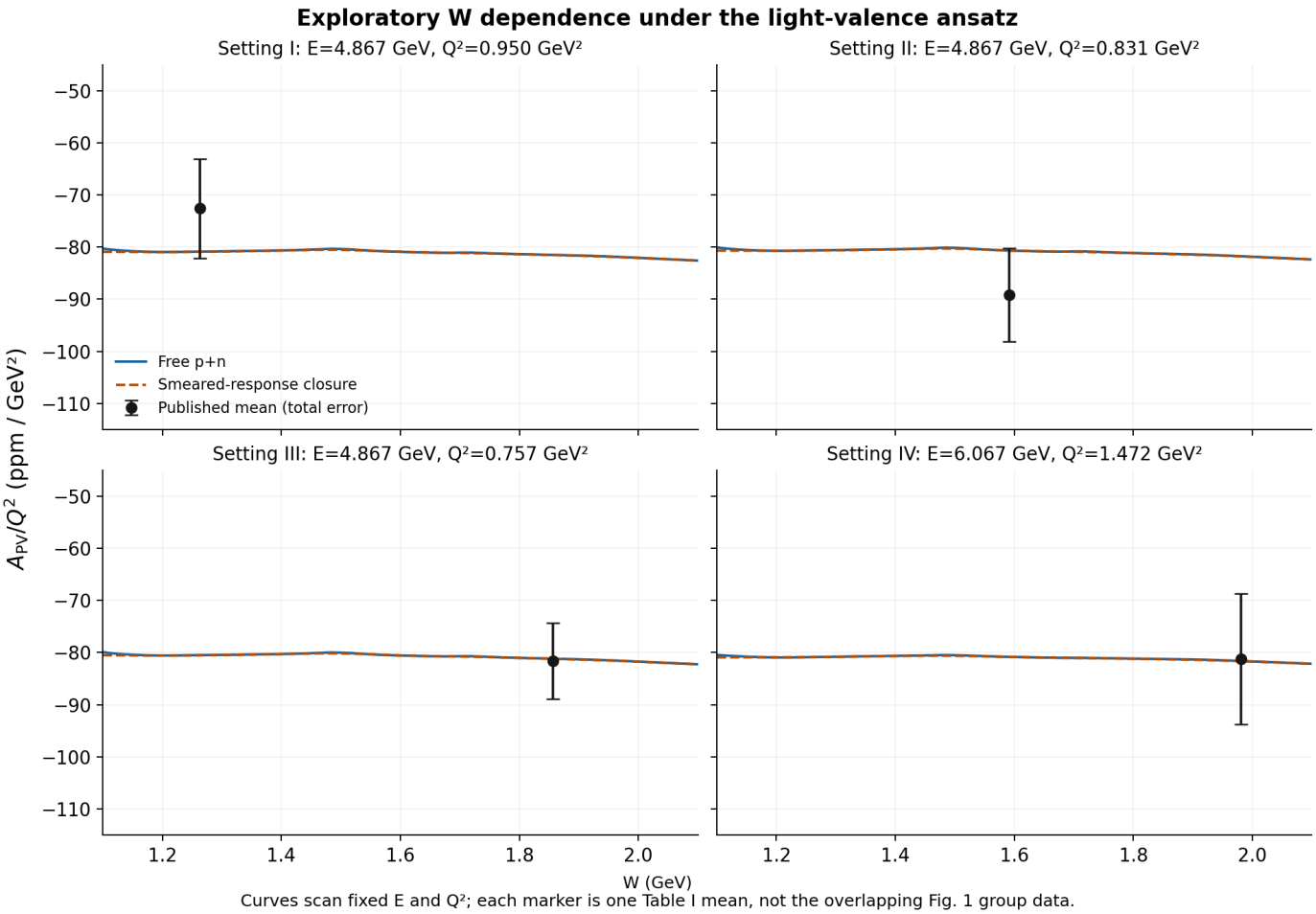
The effective flavor inversion is U=6(4F1p-F1n)/5 and D=6(4F1n-F1p)/5. In the isoscalar sum, transverse resonance shapes cancel from the ratios F1gammaZ/F1gamma=(6/5)(2gVu-gVd) and F3gammaZ/F1gamma=18/5. These ratios are assumptions of the effective light-valence construction, not extracted literal resonance PDFs.

```
r2 = 1 + 4*M*M*x*x/Q2
R  = r2*F2/(2*x*F1) - 1
D  = 1 + (1-y)^2 - y*y*(1-r2/(1+R)) - x*y*M/E
Y3 = r2/(1+R) * y*(2-y)/D
A_ppm = 3*GF*Q2/(10*sqrt(2)*pi*alpha)*1e6*(C1*Y1+C2*Y3)
```

Nominally RgammaZ=Rgamma and Y1=1. The finite-mass term follows Eq. (13) of arXiv:0801.4791v4 [2], and is checked by a direct response contraction. This study uses the half-isospin coupling convention explicitly listed above, rather than importing differently normalized couplings from another formula.

The smeared alternative imposes the same transverse response on F1gamma, F1gammaZ and F3gammaZ. This is a common-response closure, not a calculated weak axial smearing model. No unsmeared weak numerator is divided by a smeared electromagnetic denominator.

Exploratory dependence on W



Curves scan fixed E and Q²; each marker is one Table I mean, not the overlapping Fig. 1 group data.

Setting	Grid W range (GeV)	Rows	Free A range (ppm)
I	1.100 to 2.100	201	-78.466 to -76.287
II	1.100 to 2.100	201	-68.456 to -66.533
III	1.100 to 2.100	201	-62.258 to -60.493
IV	1.100 to 2.100	201	-120.920 to -118.420

Each curve fixes E and Q2 and varies W. Each data point is one published Table I mean. This does not reconstruct the overlapping group-trigger bins of the paper's Fig. 1. Smooth behavior partly follows from the imposed isoscalar flavor cancellation and must not be presented as independent evidence that weak resonance structure is absent.

Input variations and interpretive limits

These are controlled changes of assumptions, not a model uncertainty estimate or an uncertainty budget. The nominal value is the common-response closure.

Setting	Nominal A	s2=0.232 shift	Rgz=0.8R shift	Rgz=1.2R shift
I	-76.857	-1.710	+2.082	-2.082
II	-67.018	-1.454	+2.930	-2.930
III	-61.463	-1.407	+2.671	-2.671
IV	-120.195	-2.871	+5.022	-5.022

All entries are in ppm. In the RgammaZ variations, the transverse flavor prescription and electromagnetic response are held fixed; only the weak longitudinal ratio changes. The sin2(thetaW) variation changes the declared effective couplings. Neither variation is tuned to the data.

Interpretation	Boundary
Supported	The specified model construction can be executed reproducibly and compared numerically with four measured means.
Not supported	A precision weak/nuclear model, pointwise duality, acceptance or radiative validation, a rate forecast, or a fitted coupling ellipse.
Physical limitations	Sea quarks and axial resonance amplitudes are absent. Common weak/nuclear response is imposed. Published corrected CJ theory is not reproduced.
Scheduling limitation	A 30-day total budget is recorded. No beam-time-dependent observable was computed in this benchmark.

Current missing inputs

Input	Status / consequence
Colab notebook and fit inputs	notebook_reviewed_external_tables_missing: HT/CSV templates and fit implementation are reviewed; 34 referenced data files and clean-run fixes are still needed.
Original READPDF3 and MMHT grid	not_supplied: Inherited PDF calculation cannot be reproduced as supplied.
Original rate files and helper routines	not_supplied: No validated statistical projection or setting optimization.
Detector and target acceptance validation	not_established_by_this_run: No detector-folded or LD2 hardware prediction.

Diagnostics and source provenance

11 collection gates passed. Maximum absolute discrepancy when reconstructing saved numeric CSV columns from raw values: 0.000e+00. This checks pipeline consistency, not independent physical correctness.

Gate	Status
pinned model blob	PASS
pinned table blobs	PASS
published data fingerprint	PASS
generation input and output hashes	PASS
raw output hash	PASS
byte identical fresh process repeat	PASS
unique and complete point ids	PASS
all 808 points reconstructed from raw	PASS
all csv numeric columns consistent	PASS
Fortran invalid input rejection	PASS
missing table rejection	PASS

Model commit: 2a27ddb7442ab2148e47fe4f3e998cf1be0a9fa8

Model Git blob: 02a1cea8a4850352d4c1f924de3009db2ede7aef

Table repository commit: 641b8c07182fd045fd8a5c007cdc1b71a418f3ce

Table	SHA-256
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f12-onshell-cdbonn.dat	7fa11ccbf2beab847f26bb3ba1c2ee7116e50646b9ee12746f200d7ad1ac42e0
f22-onshell-cdbonn.dat	6720800a1c3216e8938987f3e064e0c28d0101fdb81c1fc457c00580dab47f46

Snapshot manifest SHA-256: 3031e8792efd49b2e59b568530350dca9e677dadfbfe9b0c294ae86b805f4015

The complete manifest also pins code, raw output, published data, diagnostic JSON, logs, and figures. Before each report build, every indexed snapshot is checked against its manifest. Earlier files are never updated by the collection workflow.

Run results: Reproducible resonance benchmark - report baseline

Run ID: 20261025T181024773791Z

Collection mode: full_fortran_run. This collection did rerun Fortran. The original raw values are retained and independently checked against the saved analysis.

Setting	E (GeV)	Q2 (GeV2)	W (GeV)	Data A (ppm)	Stat.	Syst.	Total
I	4.867	0.950	1.263	-68.97	8.47	3.30	9.09
II	4.867	0.831	1.591	-74.12	6.87	2.84	7.43
III	4.867	0.757	1.857	-61.80	5.08	2.11	5.50
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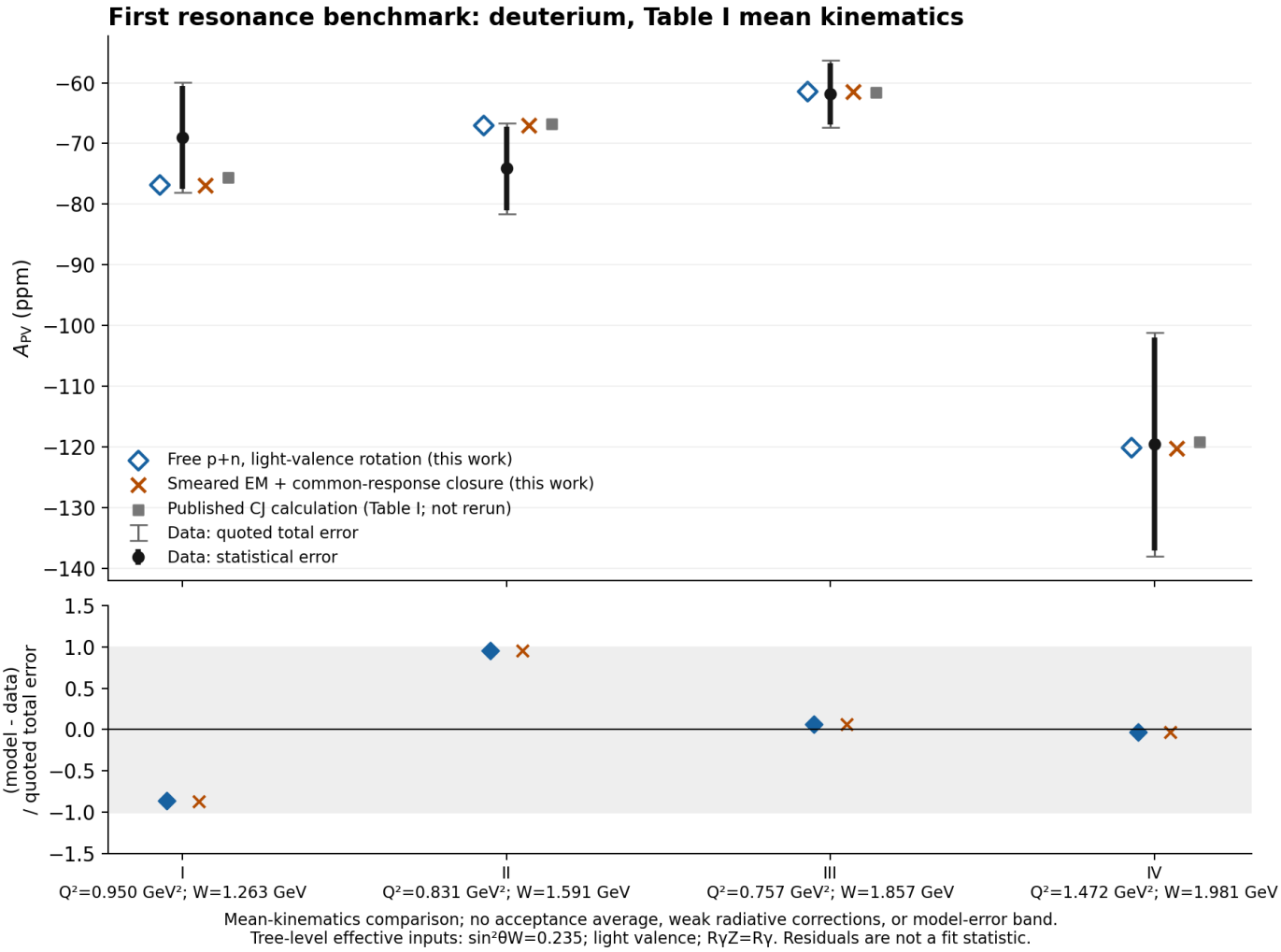
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I	-76.836	-76.857	-7.887	-0.868
II	-67.039	-67.018	7.102	0.956
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IV	0.32595	0.39666	14.7921	0.44465	0.44913

Setting	F1 p+n	F1 d	F1 change (%)	R free	R d
I	0.561875	0.531351	-5.433	0.18034	0.17268
II	0.733047	0.769094	4.917	0.30753	0.31790
III	1.116983	1.127500	0.942	0.34404	0.33441
IV	0.727866	0.724939	-0.402	0.34534	0.33028

Setting	Coefficient C1 (ppm)	Coefficient C2 (ppm)	Closure - free (ppm)
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sin2(thetaW)	0.235; retained effective tree-level baseline
C1; C2	2C1u-C1d = -0.7166667; 2C2u-C2d = -0.09
Couplings	gAe=-1/2; gVe=-1/2+2s2; gVu=1/2-4s2/3; gVd=-1/2+2s2/3; Ciq=2*g_e*g_q
Flavor assumption	Light valence only; charge symmetry; qminus=qplus; no independently supplied sea or axial resonance amplitudes.

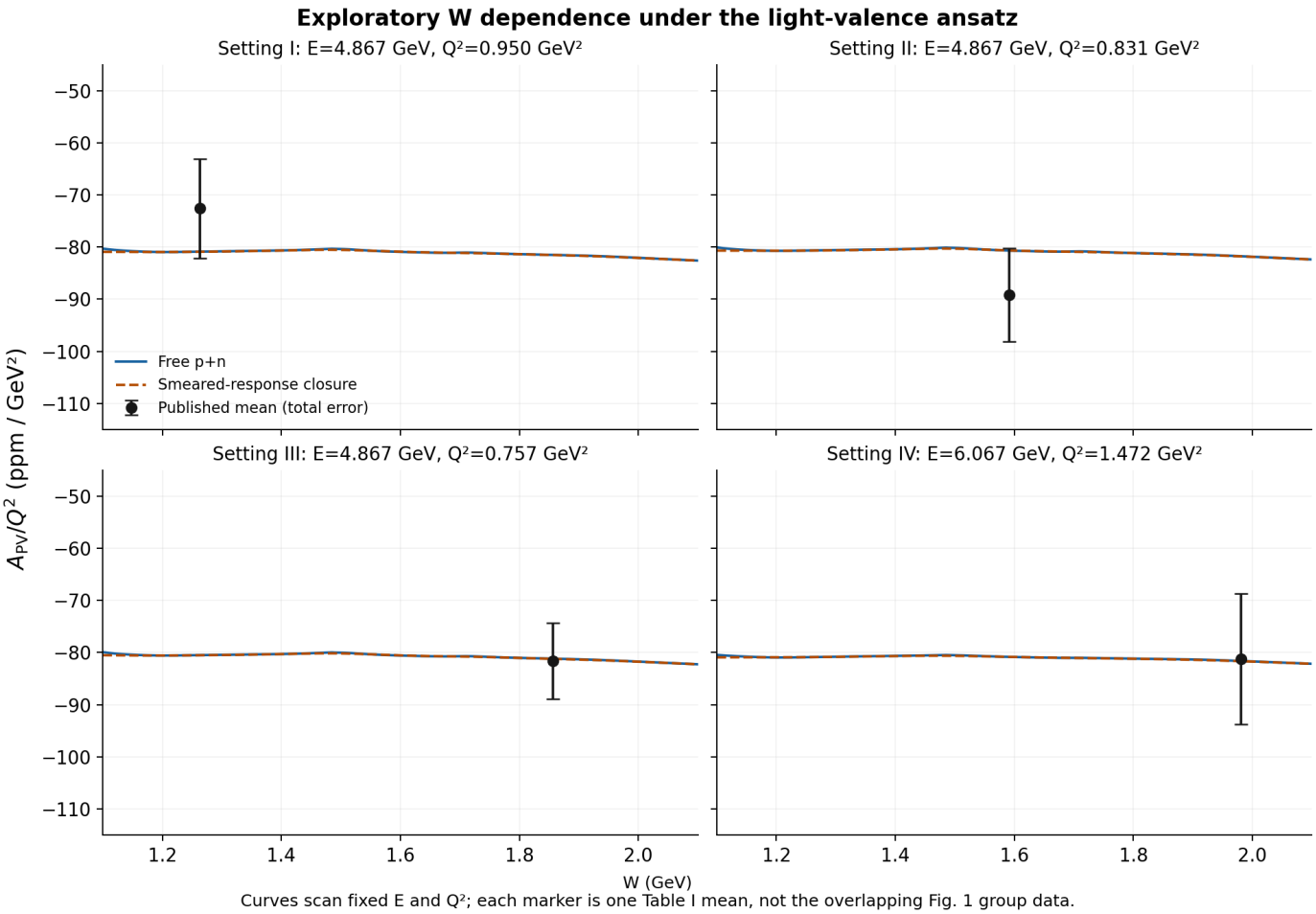
The effective flavor inversion is U=6(4F1p-F1n)/5 and D=6(4F1n-F1p)/5. In the isoscalar sum, transverse resonance shapes cancel from the ratios F1gammaZ/F1gamma=(6/5)(2gVu-gVd) and F3gammaZ/F1gamma=18/5. These ratios are assumptions of the effective light-valence construction, not extracted literal resonance PDFs.

```
r2 = 1 + 4*M*M*x*x/Q2
R  = r2*F2/(2*x*F1) - 1
D  = 1 + (1-y)^2 - y*y*(1-r2/(1+R)) - x*y*M/E
Y3 = r2/(1+R) * y*(2-y)/D
A_ppm = 3*GF*Q2/(10*sqrt(2)*pi*alpha)*1e6*(C1*Y1+C2*Y3)
```

Nominally RgammaZ=Rgamma and Y1=1. The finite-mass term follows Eq. (13) of arXiv:0801.4791v4 [2], and is checked by a direct response contraction. This study uses the half-isospin coupling convention explicitly listed above, rather than importing differently normalized couplings from another formula.

The smeared alternative imposes the same transverse response on F1gamma, F1gammaZ and F3gammaZ. This is a common-response closure, not a calculated weak axial smearing model. No unsmeared weak numerator is divided by a smeared electromagnetic denominator.

Exploratory dependence on W



Curves scan fixed E and Q²; each marker is one Table I mean, not the overlapping Fig. 1 group data.

Setting	Grid W range (GeV)	Rows	Free A range (ppm)
I	1.100 to 2.100	201	-78.466 to -76.287
II	1.100 to 2.100	201	-68.456 to -66.533
III	1.100 to 2.100	201	-62.258 to -60.493
IV	1.100 to 2.100	201	-120.920 to -118.420

Each curve fixes E and Q2 and varies W. Each data point is one published Table I mean. This does not reconstruct the overlapping group-trigger bins of the paper's Fig. 1. Smooth behavior partly follows from the imposed isoscalar flavor cancellation and must not be presented as independent evidence that weak resonance structure is absent.

Input variations and interpretive limits

These are controlled changes of assumptions, not a model uncertainty estimate or an uncertainty budget. The nominal value is the common-response closure.

Setting	Nominal A	s2=0.232 shift	Rgz=0.8R shift	Rgz=1.2R shift
I	-76.857	-1.710	+2.082	-2.082
II	-67.018	-1.454	+2.930	-2.930
III	-61.463	-1.407	+2.671	-2.671
IV	-120.195	-2.871	+5.022	-5.022

All entries are in ppm. In the RgammaZ variations, the transverse flavor prescription and electromagnetic response are held fixed; only the weak longitudinal ratio changes. The sin2(thetaW) variation changes the declared effective couplings. Neither variation is tuned to the data.

Interpretation	Boundary
Supported	The specified model construction can be executed reproducibly and compared numerically with four measured means.
Not supported	A precision weak/nuclear model, pointwise duality, acceptance or radiative validation, a rate forecast, or a fitted coupling ellipse.
Physical limitations	Sea quarks and axial resonance amplitudes are absent. Common weak/nuclear response is imposed. Published corrected CJ theory is not reproduced.
Scheduling limitation	A 30-day total budget is recorded. No beam-time-dependent observable was computed in this benchmark.

Current missing inputs

Input	Status / consequence
Colab notebook and fit inputs	notebook_reviewed_external_tables_missing: HT/CSV templates and fit implementation are reviewed; 34 referenced data files and clean-run fixes are still needed.
Original READPDF3 and MMHT grid	not_supplied: Inherited PDF calculation cannot be reproduced as supplied.
Original rate files and helper routines	not_supplied: No validated statistical projection or setting optimization.
Detector and target acceptance validation	not_established_by_this_run: No detector-folded or LD2 hardware prediction.

Diagnostics and source provenance

11 collection gates passed. Maximum absolute discrepancy when reconstructing saved numeric CSV columns from raw values: 0.000e+00. This checks pipeline consistency, not independent physical correctness.

Gate	Status
pinned model blob	PASS
pinned table blobs	PASS
published data fingerprint	PASS
generation input and output hashes	PASS
raw output hash	PASS
byte identical fresh process repeat	PASS
unique and complete point ids	PASS
all 808 points reconstructed from raw	PASS
all csv numeric columns consistent	PASS
Fortran invalid input rejection	PASS
missing table rejection	PASS

Model commit: 2a27ddb7442ab2148e47fe4f3e998cf1be0a9fa8

Model Git blob: 02a1cea8a4850352d4c1f924de3009db2ede7aef

Table repository commit: 641b8c07182fd045fd8a5c007cdc1b71a418f3ce

Table	SHA-256
f11-onshell-cdbonn.dat	da9681af403662b06df641d48c18fb3903d11d5f257e28d7f486d7d10567c9e9
f12-onshell-cdbonn.dat	7fa11ccbf2beab847f26bb3ba1c2ee7116e50646b9ee12746f200d7ad1ac42e0
f22-onshell-cdbonn.dat	6720800a1c3216e8938987f3e064e0c28d0101fdb81c1fc457c00580dab47f46

Snapshot manifest SHA-256: e226720eccaf0bf6476461e2a27c4d473585275cda4b3fa6b057a4cf5e969c52

The complete manifest also pins code, raw output, published data, diagnostic JSON, logs, and figures. Before each report build, every indexed snapshot is checked against its manifest. Earlier files are never updated by the collection workflow.

Negative controls on temporary copies

Deliberately invalid evidence	Outcome
stale csv	PASS: rejected
resealed wrong number	PASS: rejected
changed source	PASS: rejected

These controls verify rejection of stale CSVs, changed source, and a wrong numerical value even after its output hash is resealed. Original snapshots are checked again afterward.

Fit notebook: received evidence and scope

Review ID: notebook_afc987aefb60_audit_3f44c035

The supplied notebook has 39 cells. Its byte-exact source, exported cell text, saved-output inventory, inline data, numerical probes, and findings are preserved in a separate sealed notebook-review folder. Cell references below are zero-based.

Evidence	Status
Uploaded notebook	Received; source SHA-256: afc987aefb602b4310b0b8e71aa3274ddb634ac37d522d912a736117a79e8193
Full notebook run	Not performed. Drive was not mounted, and external projection files were not loaded.
Selected numerical check	Only reviewed function definitions from cells 6, 14, 15 and the inline Hall A DIS dataset assignment from cell 5 were executed.
External data	34 files referenced: 28 Hall C setting tables, four HRS tables, solid11.dat, and pvdis_11gev_CJ15nlo.csv. Not supplied in the notebook.
Cached output	Historical evidence only. All execution_count fields are null; retained outputs do not establish a clean, ordered execution.
Clean-run blockers	Undefined setting groups and HallC1 result variables are additional blockers even if external files become available.

Hadronic templates are now explicit

```
A_i = (c1_i*X + c2_i*Y) * (1 + beta_HT/((1-x_i)^3*Q2_i)
                               + beta_CSV*x_i^2)
X = 2*C1u - C1d;  Y = 2*C2u - C2d
beta_HT: GeV^2; beta_CSV: dimensionless
```

None means a free nuisance parameter; numeric values fix it. Configurations fit neither term, HT only, CSV only, or both. Both templates multiply the entire asymmetry and are applied where the second coupling coefficient is nonzero. No nuisance prior or parameter bound is supplied in the function.

This supplies an implemented ansatz, not its scientific validation. The covariance uses the X-Y submatrix of the full inverse local information matrix, which includes local nuisance marginalization when the fit is identifiable. It is not a global nonlinear profile scan.

Notebook audit: numerical findings

Check	Executed result
Joint confidence convention	Delta chi2=1 encloses 39.347% for df=2. Correct 68.3% joint threshold: 2.297707; multiply both axes by 1.515819.
Confidence argument ignored	Requests for 68.3% and 95% return identical axes because delta is overwritten with 1.0.
Hall C time scaling	At a 20% low-angle allocation, notebook factor is 0.632456; inverse-square-root-time factor relative to the same-total 50:50 reference is 1.581139. All setting files currently receive the low-angle factor.
SoLID correlation split	Notebook f=0.784585; variance fraction for the listed sources is 0.615574. Off-diagonal variance ratio: 1.274559.
Parser rejection probe	An audit-owned two-row input with one malformed numeric row returns one row with no rejection diagnostic.
Recentered parameter records	Display centers overwrite fitted params; a four-element fitted-parameter vector can become a two-element display vector while fitted/cov_full/chi2 retain the original fit.

Inline Hall A DIS implementation check

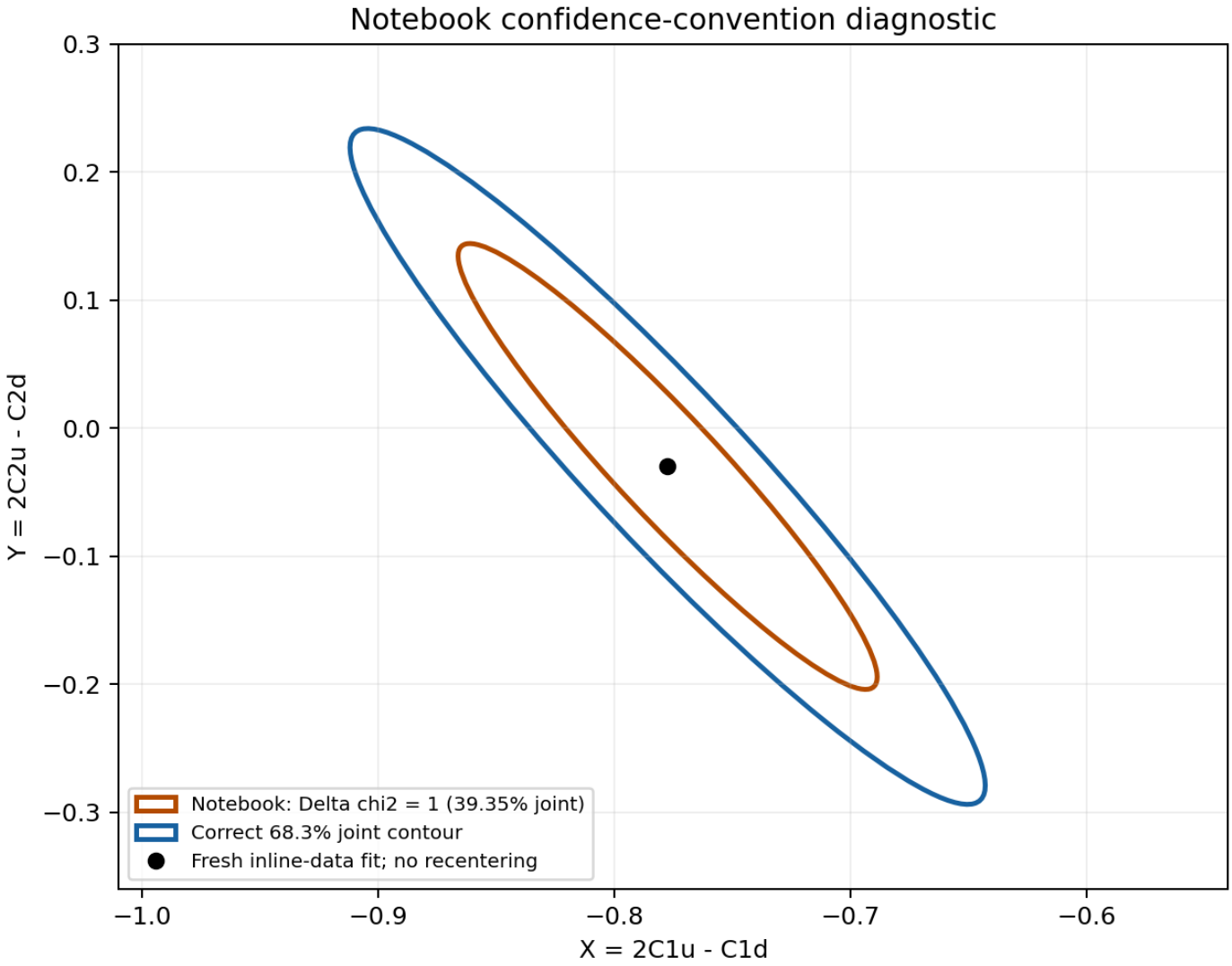
Fresh unshifted fit: X=-0.777397447, Y=-0.030058549; sigma_X=0.088731016, sigma_Y=0.174050281; correlation=-0.944290.

Independent GLS reproduces this center and covariance. The nonlinear function with both betas fixed at zero reproduces the same result. The fresh center agrees with the saved printed notebook center to its displayed rounding. These are checks of the inline implementation, not an independently validated world fit. These two DIS points are separate from the four resonance means earlier in this report.

Thirty-day budget

The time-factor comparison above is conditional on a same-total 50:50 reference. It is not itself a new 30-day projection. If inherited errors are for 30 days per setting, each must instead be scaled by sqrt(30 days / actual allocated time). An explicit schedule must partition the user's 30 days and account for simultaneous spectrometers.

Confidence-convention diagnostic



Only the two inline Hall A DIS measurements and the notebook's within-dataset covariance were used here. Both contours share the fresh unshifted fit center. The larger contour uses the correct joint 68.3% threshold; the smaller reproduces the notebook's Delta chi2=1 convention. No Hall C or SoLID projection was rerun.

A covariance ellipse with Delta chi2=1 can be reported as that convention, with its projections equal to one-standard-deviation parameter errors. It must not be labeled a joint 68.3% confidence region in two dimensions.

Notebook findings and prioritized corrections

ID / cells	Finding and correction
N01 / 14,15	Joint confidence threshold overwritten. Expose delta-chi2/coverage explicitly and retain separate one-parameter errors and joint contours.
N02 / 3,8	Hall C time scaling is inverted and applied to every setting. Use sqrt(reference_time/allocated_time) from explicit setting-pair exposure records; reject zero exposure and duplicate time allocations.
N03 / 4,14,15	Amplitude correlation fraction used as variance fraction. Define correlated source amplitudes directly or use the squared amplitude fraction consistently; document the intended decomposition.
N04 / 22,28	Undefined setting groups prevent a clean sequential run. Define the setting groups from an explicit schedule and remove dependencies on hidden interactive state.
N05 / 18,19,20,23,30	Display recentering overwrites fitted parameter records. Preserve fitted_params and covariances; store display_center separately, with the shift and purpose recorded.
N06 / 14,15	Common systematics stop at dataset boundaries. Represent each physical systematic as a source vector spanning every affected measurement; validate signs and overlap.
N07 / 5,15	Hadronic templates supplied, but application and authority need review. Retain this exact ansatz as notebook provenance; validate its scientific interpretation and use an explicit apply_hadronic flag before the PAC fit.
N08 / 5,18	Derived external constraints treated as independent priors. Recover the upstream likelihood/provenance and shared information; do not claim an independently combined world fit until this is resolved.
N09 / 6,8,11,12	External tables missing and parsers silently skip rows. Supply exact tables and hashes; use explicit schemas, row rejection logs, unit checks, and finite-value/rate validation.
N10 / 8,30,36	Labeling and legacy contour geometry need cleanup. Correct labels and retain historical contour points with their original confidence convention and provenance.

Next inputs and reproduction

Obtain the 34 named files in external_inputs.csv with their source hashes and exposure definitions. Resolve confidence labels, inverse time scaling, systematic-source covariance and missing setting definitions before accepting Hall C sensitivity rankings. Verify external-prior provenance and possible shared information before claiming a combined fit.

```
python tools/verify_notebook_review.py --recompute
python tools/build_study.py --mode report
```

The notebook-review manifest pins original source, harness, numerical outputs and plots separately from the resonance-run registry. No original notebook or previous resonance snapshot was modified.

Machine-readable evidence and column dictionary

Each runs/<run_id>/ directory is a portable snapshot. diagnostics.json includes derived responses at all 808 points; results/*.csv retain the data tables; run.json records scope and capture mode; manifest.json records SHA-256 and byte count for each artifact. JSON uses finite numeric values and schema-version labels.

Measured inputs, identifiers, and kinematics

Column	Unit	Definition
setting	label	Published setting I-IV
E_GeV	GeV	Incident beam energy
Q2_GeV2	GeV^2	Positive squared four-momentum transfer
W_GeV	GeV	Hadronic mass using nucleon convention
A_phys_ppm	ppm	Published physics asymmetry after the experiment's corrections; not the raw helicity asymmetry.
stat_ppm	ppm	Published one-standard-deviation statistical error on A_phys.
syst_ppm	ppm	Published systematic error on A_phys; correlation matrix is unavailable here.
total_reported_ppm	ppm	Published total experimental error; its rounded quadrature is checked against stat and syst.
CJ_published_ppm	ppm	CJ-based prediction copied from the paper for comparison; not recomputed by this study.
point_id	integer	1-4: published means; 1000*s+j: grid, s=1..4, j=0..200
x	dimensionless	Bjorken x = Q2/[W2-M2+Q2], using the declared nucleon mass.
y	dimensionless	Fractional electron energy loss: nu/E, with nu=(W2-M2+Q2)/(2M).
theta_deg	degree	Angle implied by mean Q2,W,E; not central spectrometer angle

For the scan, setting is encoded in point_id (1000*s+j, s=1..4, j=0..200); 1-4 identify the measured means. A_residual fields are present only for those means. The machine dictionary is retained in each snapshot.

Column dictionary: model responses and variations

Column	Unit	Definition
F1p	dimensionless	Computed free-proton transverse electromagnetic structure function.
F1n	dimensionless	Computed free-neutron transverse electromagnetic structure function.
F1_free_sum	dimensionless	F1p+F1n; electromagnetic reference for the free p+n construction.
F1d_smeared	dimensionless	Per-deuteron F1 from the CD-Bonn on-shell electromagnetic smearing model; QE excluded.
F2_free_sum	dimensionless	Computed electromagnetic F2p+F2n.
F2d_smeared	dimensionless	Per-deuteron electromagnetic F2 after the specified smearing; QE excluded.
R_free	dimensionless	Electromagnetic longitudinal/transverse ratio from the free p+n F1 and F2.
R_smeared	dimensionless	Electromagnetic longitudinal/transverse ratio from the smeared deuteron F1 and F2.
Y3_free	dimensionless	Finite-Q2 axial kinematic factor using R_free.
Y3_closure	dimensionless	Finite-Q2 axial kinematic factor using R_smeared in the common-response ansatz.
A_free_ppm	ppm	Calculated free-p+n asymmetry under charge symmetry, light valence, and RgammaZ=Rgamma.
A_closure_ppm	ppm	Calculated asymmetry imposing a common transverse weak/electromagnetic smeared response.
A_closure_SW2_0232_ppm	ppm	Closure asymmetry with sin2(thetaW)=0.232; an input convention variation, not a fitted value.
A_Rgz_08R_ppm	ppm	Closure asymmetry with RgammaZ=0.8*Rgamma; illustrative scenario, not an error endpoint.
A_Rgz_12R_ppm	ppm	Closure asymmetry with RgammaZ=1.2*Rgamma; illustrative scenario, not an error endpoint.
coeff_C1_closure_ppm	ppm	Signed coefficient multiplying dimensionless 2C1u-C1d in the closure asymmetry.
coeff_C2_closure_ppm	ppm	Signed coefficient multiplying dimensionless 2C2u-C2d in the closure asymmetry.
residual_closure_ppm	ppm	Closure-model asymmetry minus the published physics asymmetry.
residual_over_total	dimensionless	(closure model - measured asymmetry)/quoted total error; not a fit statistic

These explanatory definitions use data/column_descriptions.json, whose hash is recorded with this PDF build. They do not modify any numeric value or historical snapshot. Coefficients are in ppm per unit dimensionless coupling combination.

Reproduction and debugging

The default build collects verified retained evidence and rebuilds the cumulative report; it does not rerun the Fortran model. Use refresh to re-execute analysis and plots from retained raw output, or full to compile and rerun the model. Each successful collection adds a new snapshot.

```
# From the study directory (Python with analysis dependencies):
python tools/build_study.py --mode collect
python tools/build_study.py --mode refresh
python tools/build_study.py --mode full

# Verify snapshots and regenerate PDF without adding a run:
python tools/build_study.py --mode report

# If reportlab is in a different Python installation:
python tools/build_study.py --mode report --report-python <python>

# Verify a snapshot from its own directory:
python tools/verify_snapshot.py <snapshot-directory>
```

Install the pinned analysis and reporting requirements into appropriate environments. A single environment can be used if compatible package versions are resolved and recorded; the captured run used separate analysis and PDF interpreters. Full mode requires WSL on Windows, GCC-13 tools and the workspace compiler, or PVDIS_FC/PVDIS_RUNTIME_LIB overrides on Linux.

Failure	First check
stale generation	A source/input/output hash changed. Refresh only for analysis changes; use full for model/driver/build changes.
model/table mismatch	Check exact Git blob identity and table commit. No replacement table is accepted silently.
CSV reconstruction mismatch	Inspect point_id, raw sf_values.dat row, data_dictionary.json, and diagnostics for the affected row.
model execution fails	Read build.log, command logs, compiler version, and rejected-input logs. Model FP traps and bounds checks are enabled.
snapshot changed	Restore exact original snapshot bytes from the review bundle; do not rehash a changed snapshot as if it were the original.

Build metadata and the report's content hash are recorded beside the PDF. The review ZIP contains the PDF, all indexed snapshots, current tools, raw data, tables, and manifests; compiler package binaries are excluded to keep it portable. No commit, push, or hosted CI is part of this workflow.

Sources and continuing study record

[1] Wang et al., resonance measurements

https://arxiv.org/pdf/1304.7741v2

Table I measurements, uncertainties, and published CJ reference values; Eq. (1) sign/convention anchor.

[2] Hobbs and Melnitchouk, finite-Q2 kinematics

https://arxiv.org/pdf/0801.4791v4

Eq. (13), with -x*y*M/E in the denominator. Coupling conventions are translated explicitly.

[3] Pinned F1F2IN21 model

https://github.com/trottar/mc-single-arm/blob/2a27ddb7442ab2148e47fe4f3e998cf1be0a9fa8/src/F1F2IN21_v1.0.f

Unmodified electromagnetic model source.

[4] Pinned external CD-Bonn tables

https://github.com/trottar/simc_gfortran/tree/641b8c07182fd045fd8a5c007cdc1b71a418f3ce/f1f2tables

Three exact Git blob identities independently checked against retained bytes.

[5] Gorchtein, Horowitz and Ramsey-Musolf, weak isospin rotation

https://arxiv.org/pdf/1102.3910

Appendix A after Eq. (A9): pure isospin-3/2 component ratio. Not used as a full inclusive deuteron prediction.

[6] Carlson and Rislow, resonance weak-response models

https://arxiv.org/pdf/1201.3323v2

Resonance/background rotations and helicity-amplitude limitations; context for the unresolved model-construction gate.

Future additions

Add new measurements or model runs as explicitly labeled datasets with their own provenance, assumptions, units, and validation gates. Preserve measured, modeled, projected, and fitted quantities as distinct fields. Update the run registry and regenerate this PDF from the verified snapshots.

The fit notebook has now been reviewed separately, and its implemented HT/CSV templates are recorded. The external projection tables, rate-chain validation, template physics validation, detector folding, and external-prior independence remain unresolved. New evidence must not retrospectively change the assumptions of an earlier run.